

Education

Texas A&M University

Pursuing M.S. Computer Science

College Station, TX

Expected 2028

Texas A&M University

*B.S. Computer Science & B.S. Statistics, **Summa Cum Laude***

College Station, TX

May 2026 | GPA: 3.92/4.00

Technical Skills

Data Science & Statistics: Python, R, SQL, pandas, NumPy, SciPy, scikit-learn, Bayesian inference, MCMC, hypothesis testing, FDR correction, Matplotlib

Bioinformatics & HPC: GenomeMap, PAML/codeml, dN/dS, pN/pS, FASTA pipelines, Slurm, Linux, TAMU HPRC

Machine Learning & Data Systems: PyTorch, Hugging Face Transformers, LangChain, AWS, MongoDB, Elasticsearch, Redis, Git, GitHub Actions

Programming: Java, C++, TypeScript, JavaScript, React, FastAPI, Docker, PostgreSQL

Experience

Graduate Assistant Research (GAR) - Computational Genomics

June 2026 - Present

Ioerger Lab, Texas A&M University

College Station, TX

- Investigating evidence of positive selection in *M. tuberculosis* genomes from diabetes and non-diabetes patient cohorts, analyzing approximately **4,000 genes** to identify candidate genomic differences for a manuscript in preparation.
- Built reproducible **Python** and **Slurm** pipelines for cohort QC, FASTA processing, Bayesian dN/dS estimation with **GenomeMap**, and posterior comparisons across large-scale TAMU HPRC runs.
- Validated selection signals with pN/pS enrichment, chi-square testing, Benjamini-Hochberg and local-FDR correction, drug-resistance positive controls, and independent **PAML/codeml** site-model analyses.

Software Engineering Capstone - Amazon-Sponsored

August 2025 - December 2025

Texas A&M University

College Station, TX

- Built a breach-intelligence analytics platform in a **3-person team**, implementing **Scrapy** and **Tor** ingestion plus a **LangChain** and **Gemini 2.5 Flash Lite** pipeline for entity extraction and threat classification.
- Designed storage and retrieval across **AWS S3**, **MongoDB**, **Elasticsearch**, **Redis**, and **Neptune**; delivered Streamlit and Kibana views for querying unstructured breach data by sector and geography.

Undergraduate Researcher

January 2025 - May 2025

UrbanResilience.AI Lab, Texas A&M University

College Station, TX

- Built **Python** ingestion and preprocessing pipelines integrating air-quality sensor APIs and web-scraped wildfire data across Los Angeles sources for downstream emergency-response analysis.

AI Engineering Intern

May 2024 - August 2024

Videomagic

Remote

- Developed and evaluated **PyTorch** and **Hugging Face Transformers** computer-vision workflows for AI-generated video detection, analyzing frame-level outputs and model failure modes for product deployment.
- Built **Feathers.js**, **MySQL**, and **Knex.js** APIs for model inference and data management, secured with **Auth0** and **JWT RBAC**; shipped AI features associated with **15% higher engagement**.

Projects & Competitions

PickLedgerPro | *Python, scikit-learn, pandas/NumPy, TypeScript, GitHub Actions*

- Shipped a public pick board from committed JSON: **8 team-model** and **3 player-prop** pipelines, an immutable pregame ledger, and ESPN grading every **15 minutes**.
- Score settled picks with **Brier**, **log loss**, and **ECE**. Unapproved models publish as PASS at 0 units until a version clears **100** priced settled picks. The approval list is empty.

Alpha - Tidal Hackathon 2025 | *React, TypeScript, Multi-LLM - 1st Place*

- Built a math-learning SPA orchestrating Gemini, OpenRouter, Claude, Manim, Desmos, and Wolfram Alpha; reduced tool-call failures from **25% to 6%** through output normalization and validation.